

Facilitator talking points — Generative AI in Databricks

Participant notebook: `02-lab.py` Scheduled: 11:15 AM–12:15 PM · 60 minutes

Delivery map

Time	Minutes	Mode	Surface and focus
11:15–11:20	5	Slides	Genie Code's role and the developer's validation responsibility
11:20–11:27	7	Notebook	Use Genie Code to explain the inherited governed SQL
11:27–11:39	12	Notebook exercise	Extend the promotion analysis to regional grain
11:39–11:44	5	Notebook	Run the independent governed regional checkpoint
11:44–11:54	10	Notebook exercise	Diagnose and repair the PySpark denominator bug
11:54–11:59	5	Notebook	Review <code>/optimize</code> and <code>/doc</code> proposals
11:59–12:08	9	Facilitator demo	Build and validate the unpublished AI/BI Dashboard draft
12:08–12:12	4	Optional guided activity	Improve the participant's private Genie Agent, or use as buffer
12:12–12:15	3	Notebook and discussion	Owner, risk, recovery, and practical next use

1. What Genie Code contributes

- Genie Code helps a developer understand, extend, debug, document, and optimize inherited notebook work.
- It can propose edits and execution, but the participant reviews the plan, diff, source, and result before approval.
- The developer remains responsible for the population, denominator, business meaning, permissions, and validation.

2. Understand inherited SQL before changing it

- The starting query uses the governed `fpd_metrics` Metric View.
- One row represents one promotion cohort.
- `MEASURE(...)` invokes governed measures; `GROUP BY ALL` groups by the selected dimensions.
- Multiplication by 100 changes display units, not the underlying metric definition.
- Ask Genie Code to explain the query first rather than immediately rewriting it.

3. Extend the investigation to region

- Keep `fpd_metrics` as the only source and filter to the promotion cohort.
- Group by `region_code`; retain eligible contracts, FPD5 contracts, and FPD5 rate.
- Apply the 20-eligible-contract rule at regional grain.
- The threshold stabilizes this workshop comparison; it is not part of FPD5 or a statistical-significance claim.

4. Validate generated analysis

- Generated code is a draft until an independent query produces the same rows, counts, and rates.
- The governed checkpoint should return six qualifying regions.
- `REGION_VI` is approximately **53.95%**, or **11.69 percentage points** above the **42.26%** promotion baseline.
- If results differ, inspect the source, promotion filter, grouping, measures, `HAVING` rule, denominator, and percentage conversion before continuing.

5. Diagnose the PySpark denominator bug

- Let participants observe the suspicious result: both cohorts appear to be 100%.
- The helper filters to `fpd5_flag = 1` before aggregation, removing non-FPD5 eligible contracts from the denominator.
- The repair keeps the full eligible population, counts contracts for the denominator, and sums `fpd5_flag` for the numerator.
- Validate exactly two cohorts, positive eligible counts, FPD5 counts no greater than eligible counts, and Metric View agreement within 0.05 percentage points.

6. Notebook compute versus SQL warehouse

- In this lab, Python, PySpark, `spark.sql(...)`, and `%sql` run on the notebook's Serverless compute.
- Genie Code assists with the notebook, but approved notebook cells still run on that notebook compute.
- The AI/BI dashboard and private Genie Agent execute their analytical SQL on a SQL warehouse.
- Notebook `%sql` statements should not be presented as SQL warehouse Query History activity.

7. Review optimization and documentation

- `/optimize` is a proposal, not proof that a query is faster.
- Accept a change only when the result remains identical and there is a material rationale.
- Performance claims require Query Profile evidence.
- `/doc` should capture the grouping, eligible-contract denominator, and threshold without duplicating the FPD5 formula.

8. Turn evidence into a dashboard

- The disposable dashboard uses only `fpd_metrics` and remains unpublished.
- KPI counters show the all-cohort baseline.
- The promotion filter affects the cohort comparison and store ranking, not the all-cohort KPI counters.
- Store rankings retain eligible and FPD5 counts beside the rate and require at least 10 eligible contracts.
- The point is to show that generated presentation assets still require reconciliation and publication decisions.

9. Optional private Genie Agent improvement

- Continue with the participant's private Section 01 Agent; do not create a second Agent.
- Keep `fpd_metrics` as its only data source.
- Add the smallest instruction or verified example needed for counts, observation date, investigation language, and the 10-contract ranking rule.
- Test the target question and the original cohort comparison in fresh conversations to detect regression.

Closing question

Where did AI accelerate the work, and which review or validation step prevented speed from weakening trust?